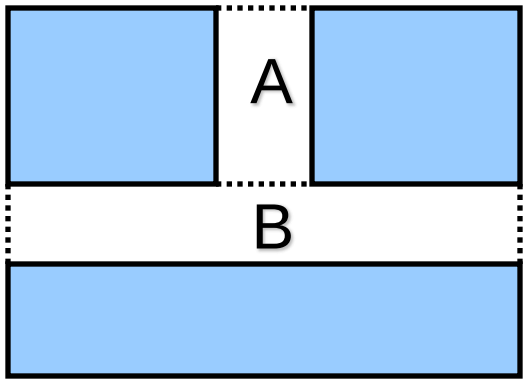
CAD for VLSI

Routing 2

After Global Routing: Detailed Routing

- The routing regions are divided into channels and switchboxes

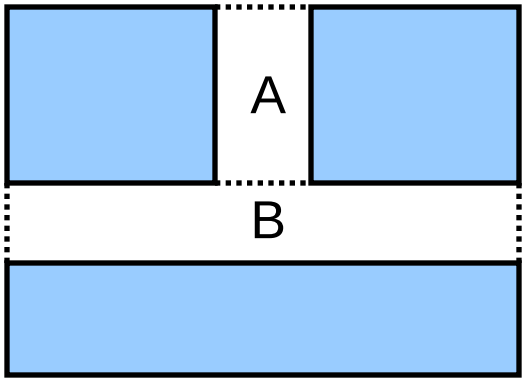


- So only need to consider the channel routing problem and the switchbox routing problem

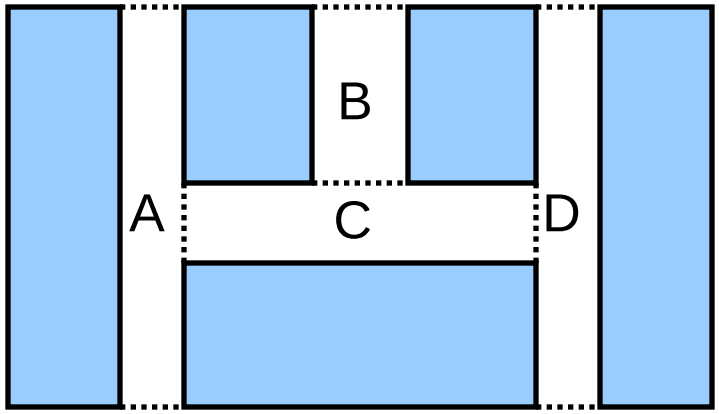
Channel Routing for Different Styles

- For gate-array design, channel widths are fixed. The goal is to finish routing of all the nets
- For standard-cell and full-custom design, channels are expandable. The goal is to route all nets using the minimum channel width
- We will consider the case when the channels are expandable

Channel Ordering



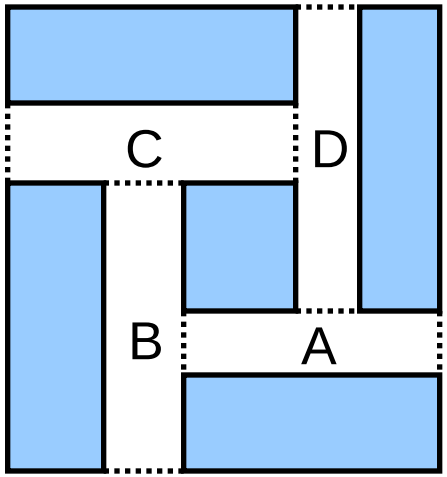
The width of A is not known until A is routed, we must route A first.



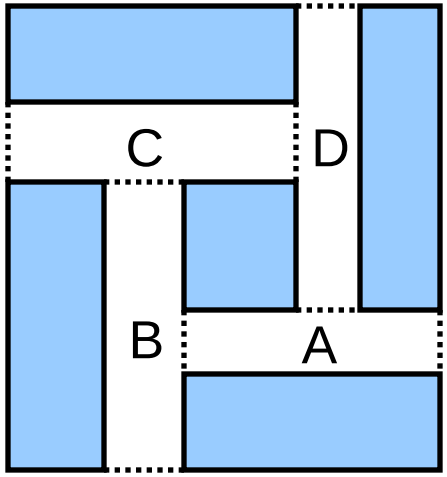
What should be the routing order for this example?

Channel Ordering

No feasible channel order!

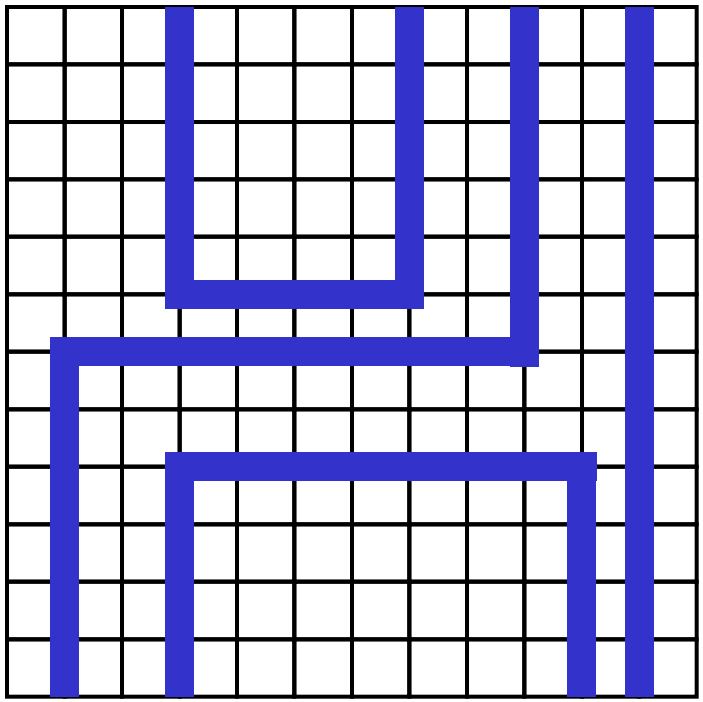


Need to use switchbox

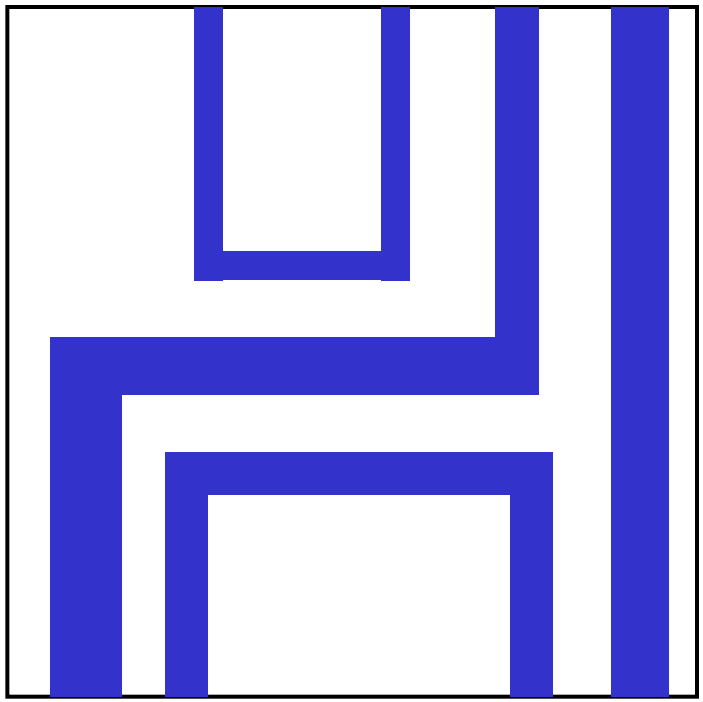


1. Fix the terminals between A & B
2. Route B, C, then D (channel)
3. Route A (switchbox)

Routing Grid Models



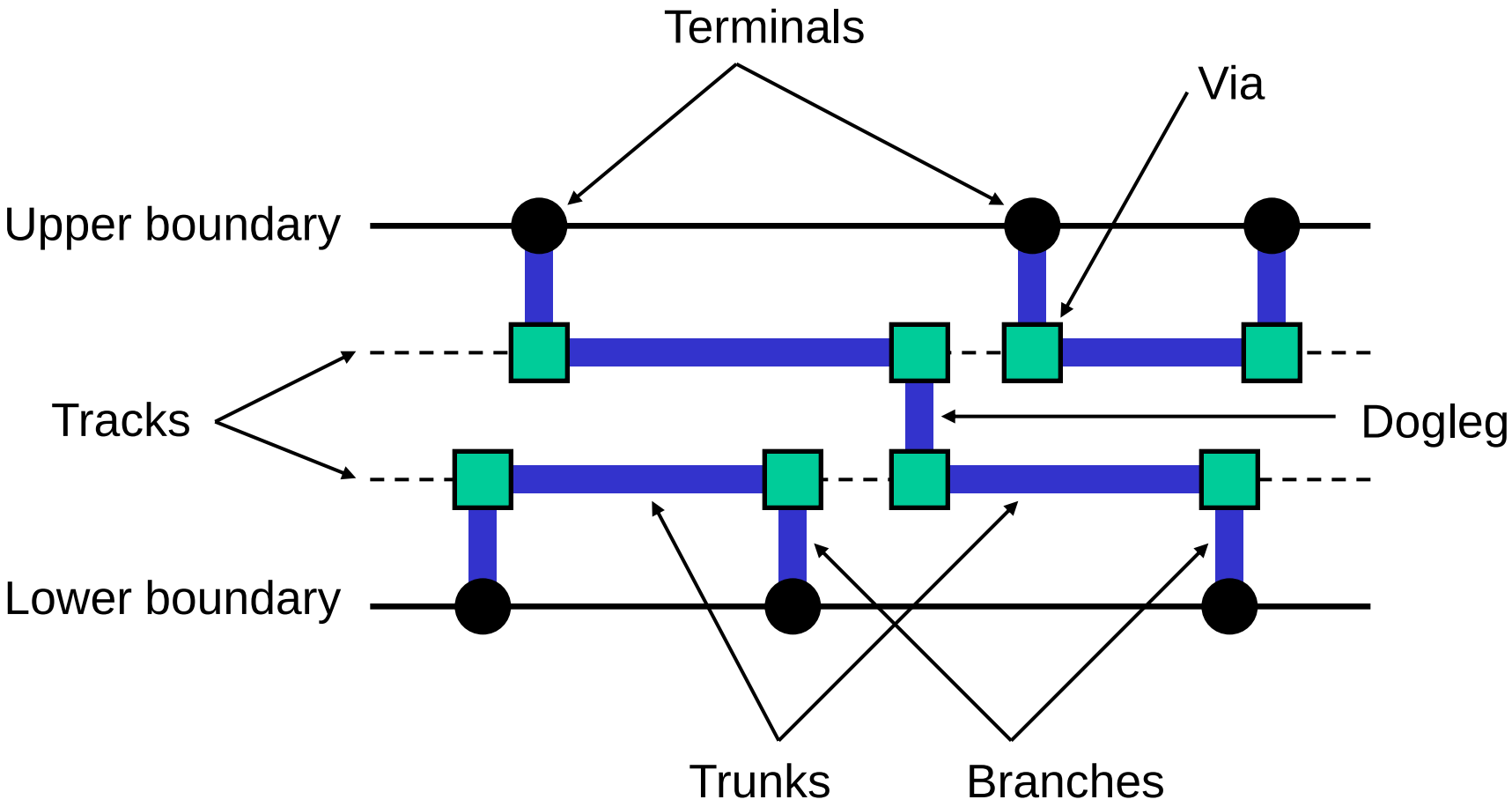
Grid-based Routing



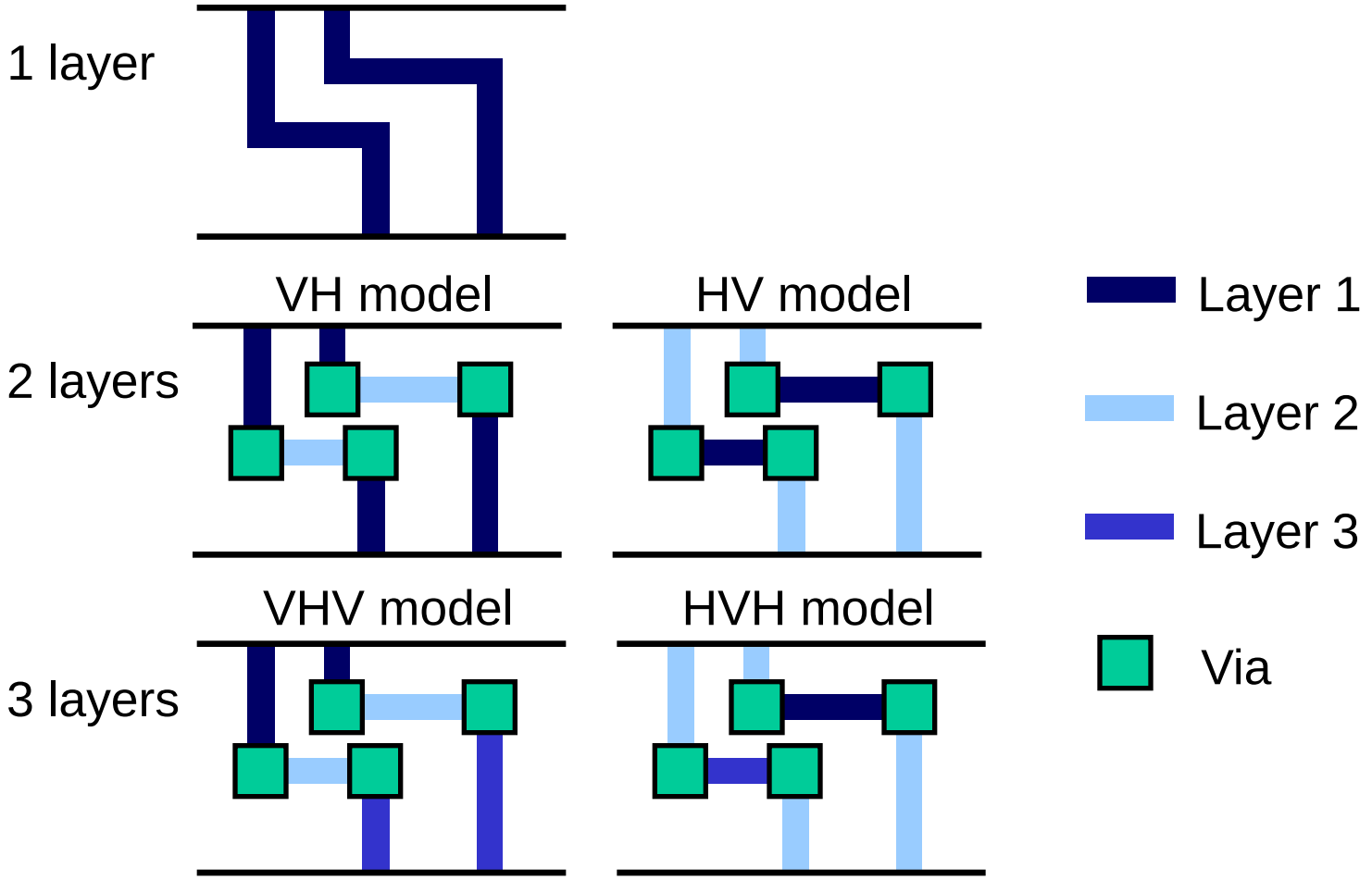
Gridless Routing

- Grid-based model is the most commonly used

Channel Routing Terminology



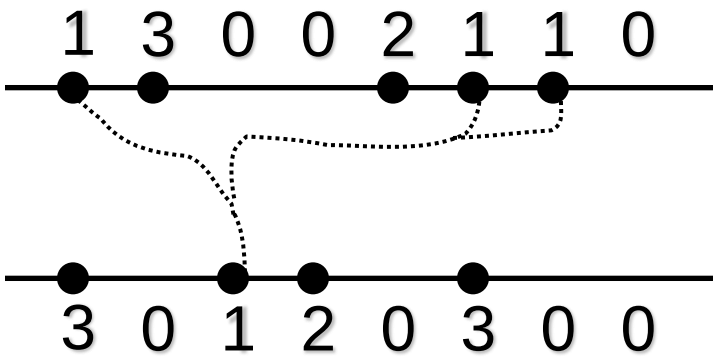
Routing Layer Models



Channel Routing Problem

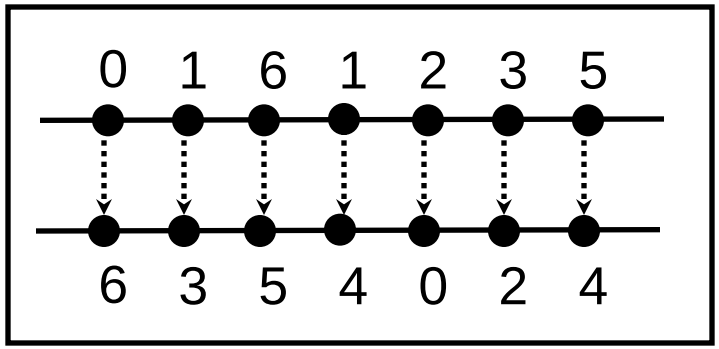
- Input:
  - Two vectors of the same length to represent the pins on two sides of the channel
  - Number of layers and layer model used
- Output:
  - Connect pins of the same net together
  - Minimize the channel width
  - Minimize the number of vias

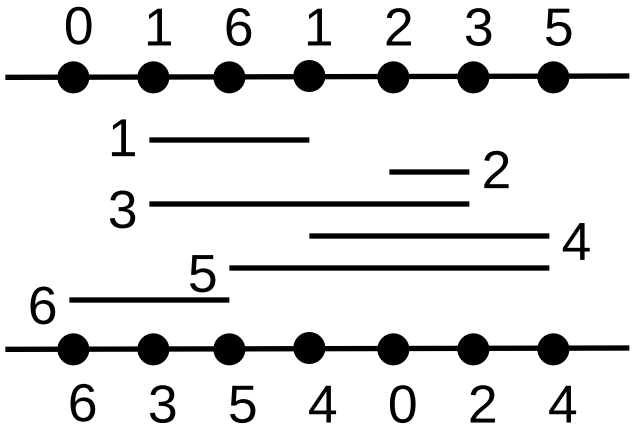
Channel Routing Problem

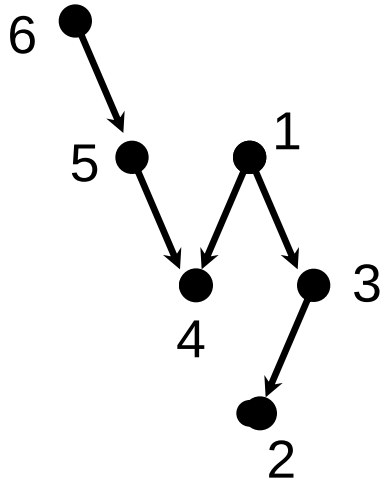


Example: (13002110)
(30120300)
where 0 = no terminal

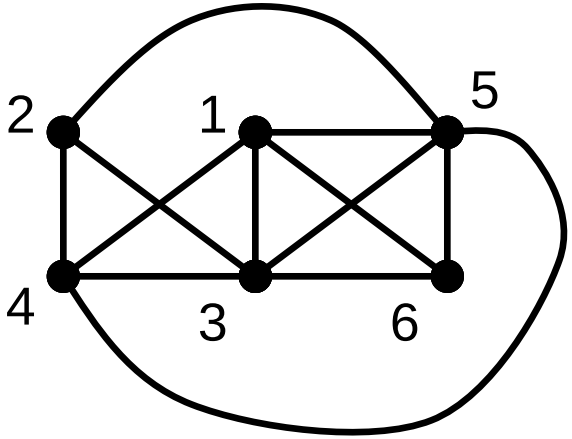
Constraint Graphs





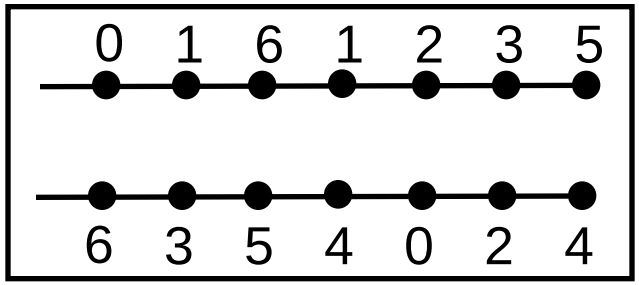


Vertical constraint graph

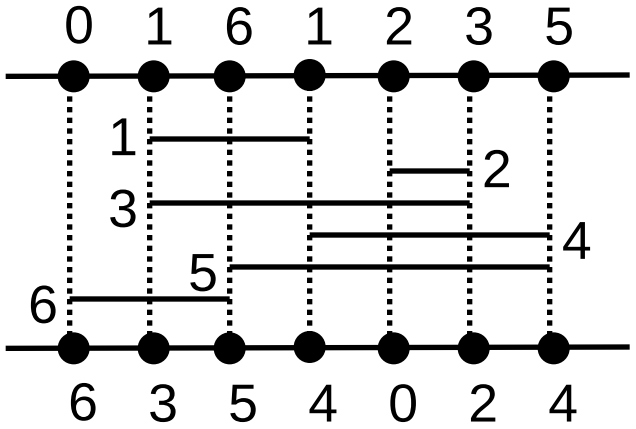


Horizontal constraint graph

Lower Bound on Channel Width



Channel density =
Maximum local density



Local density: 1 3 4 4 4 4 2

Channel density = 4

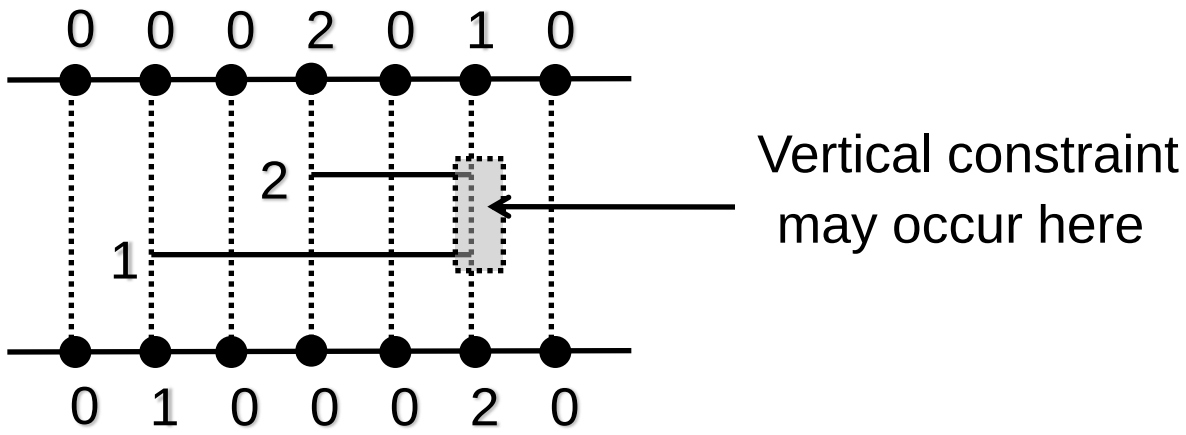
Lower bound on channel width = Channel density

Left-edge Channel Routing Algorithm

"Wire Routing by Optimizing Channel Assignment within Large Apertures", A. Hashimoto and J. Stevens, DAC 1971, pages 155-169.

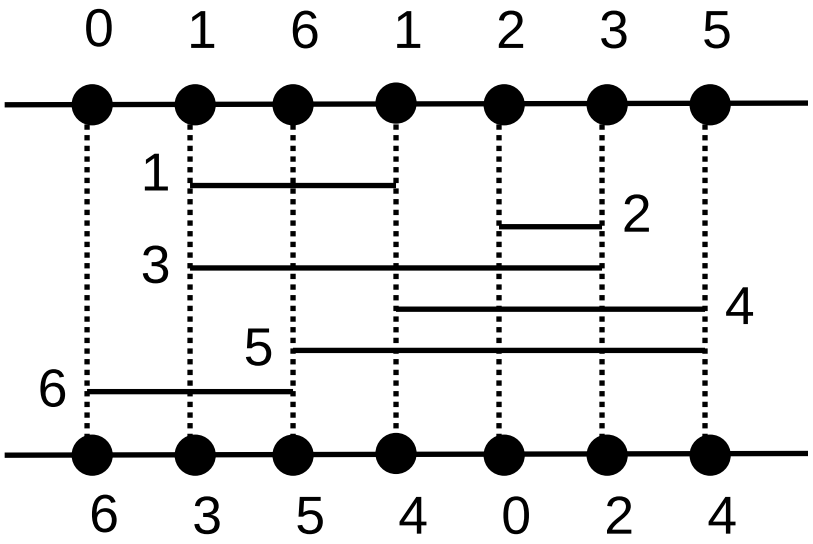
Features of Left-edge Algorithm

- Assumptions:
  - One horizontal routing layer
  - No vertical constraint, ex: VHV model

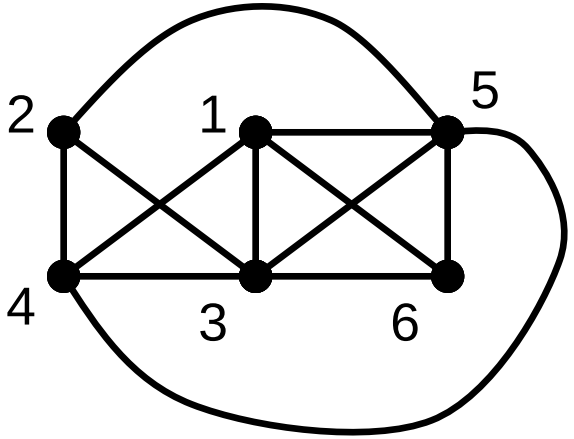


- Always gives a solution with channel width equal channel density, i.e., optimal solution

Example



Local density: 1 3 4 4 4 4 2



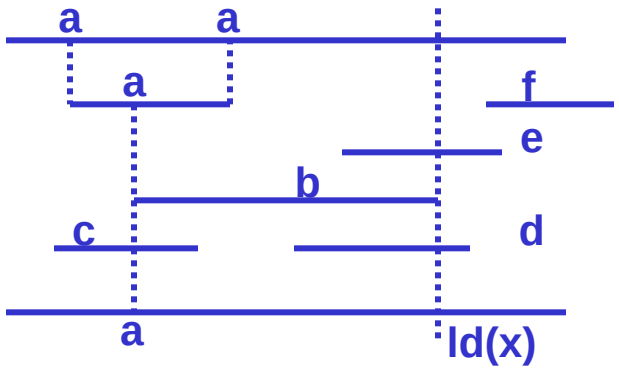
Horizontal constraint graph

Left-edge Algorithm

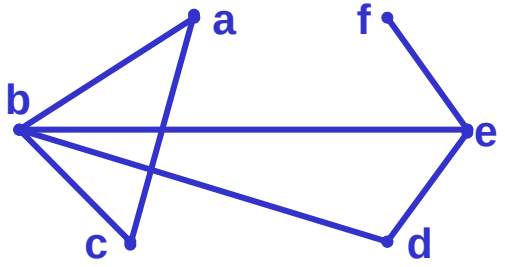
1. Sort the horizontal segments of the nets in increasing order of their left end points
2. Place them one by one greedily on the bottommost available track

Channel Density

- Local density at column C
  - ld(C) = # nets split by column C
- Channel Density
  - d = max ld(C) $\forall$ C



- Each net spans over an interval
- Horizontal Constraint Graph (HCG)
  - Node: net
  - Edge: two intervals intersect
  - Clique size = channel density
- A lower bound:
  - # tracks $\geq$ channel density



Left-Edge Algorithm for Interval Packing

Repeat

create a new track t

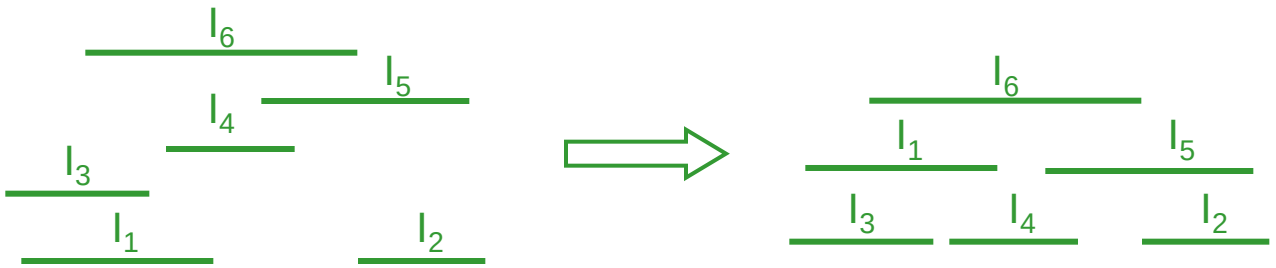
Repeat

put leftmost feasible interval to t

until no move feasible interval

until no move interval

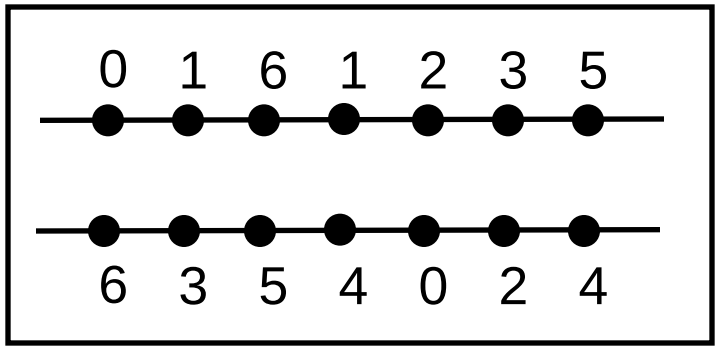
Intervals are sorted according to their left endpoints



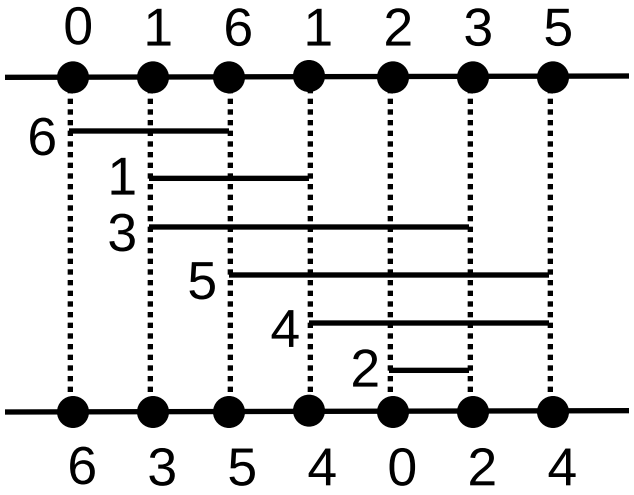
O(nlogn) time algorithm

Greedy algorithm works!

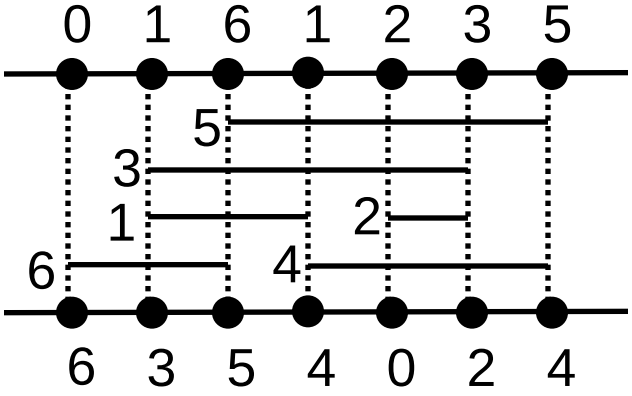
Left-edge Algorithm: Example



1. Sort by left end points

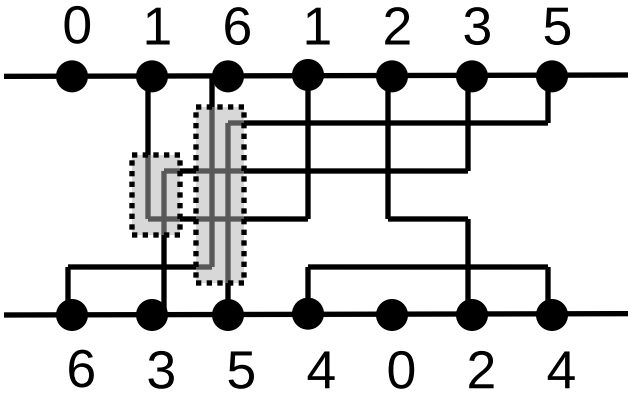


2. Place nets greedily

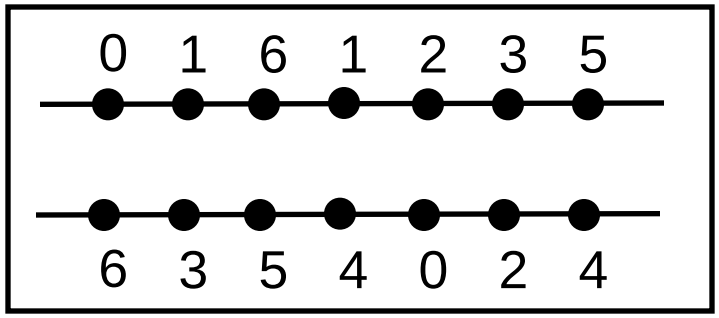


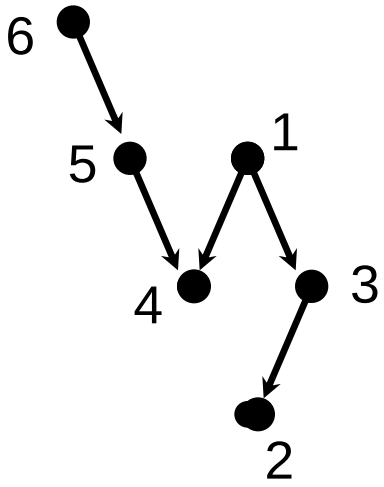
Vertical Constraint Consideration

- The Left-edge algorithm ignores vertical constraints
- When there is only one vertical layer, the algorithm will produce overlapping of vertical wire segments



Lower Bound on Channel Width





Vertical constraint graph

Length of the longest path in the vertical constraint graph

Lower bound = 3

Lower Bound on Channel Width

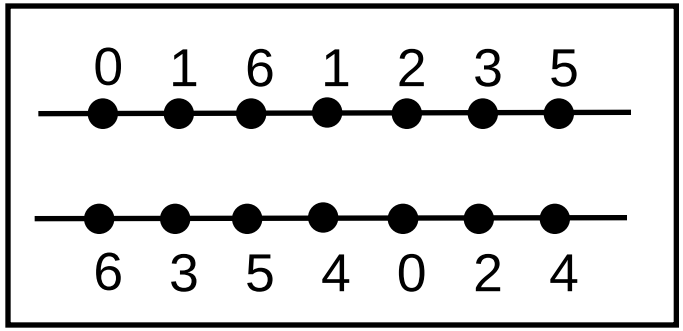
$$\text{Lower bound on channel width}$$

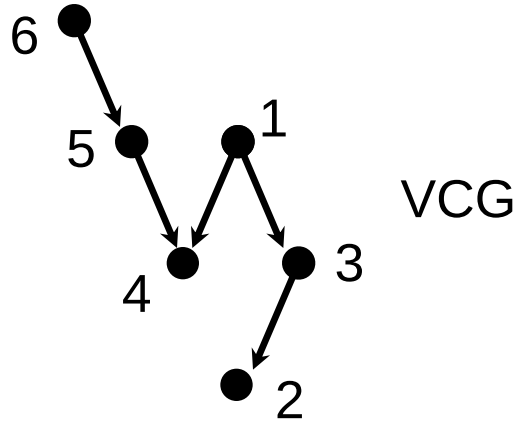
$$= \max\left\{\begin{array}{c}\text{Length of the longest path in the VCG} \\ \text{Channel density}\end{array}\right\}$$

Constrained Left-edge Algorithm

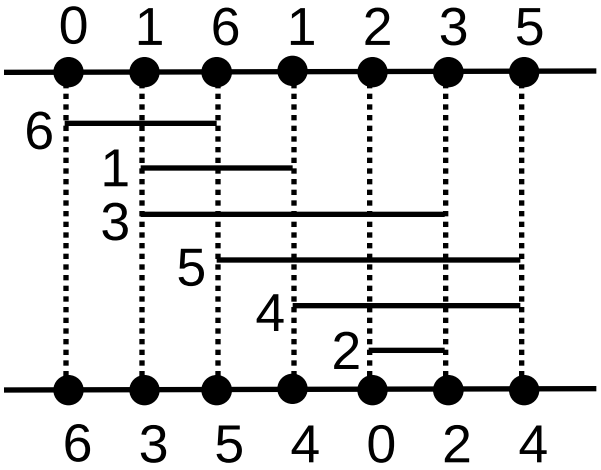
- Consider vertical constraints
- Similar to the Left-edge algorithm
- Modifications:
  - Place a horizontal segment only if it does not have any unplaced descendants in the vertical constraint graph G_v
  - Place it on the bottommost available track above all its descendents in G_v

Constrained Left-edge Algorithm

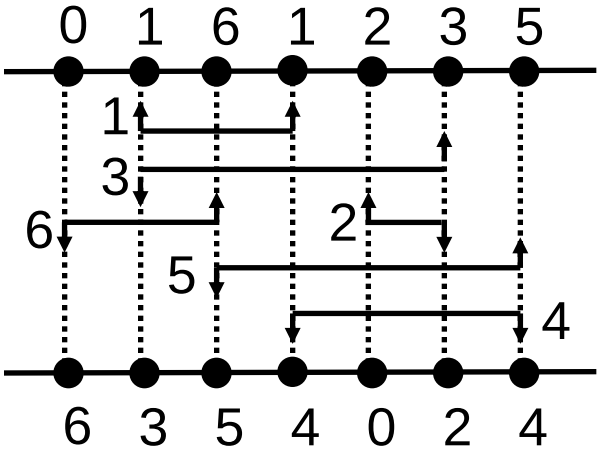




1. Sort the left end points

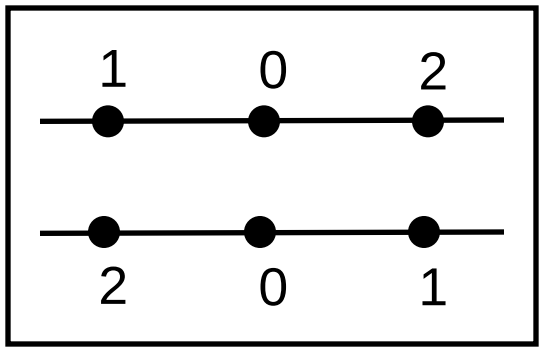


2. Place nets greedily

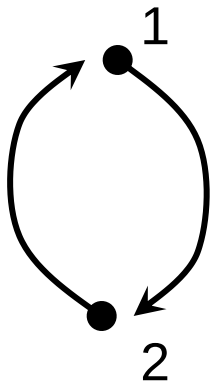


Cycles in Vertical Constraint Graph

- If there is cycle in the vertical constraint graph, the channel is not routable

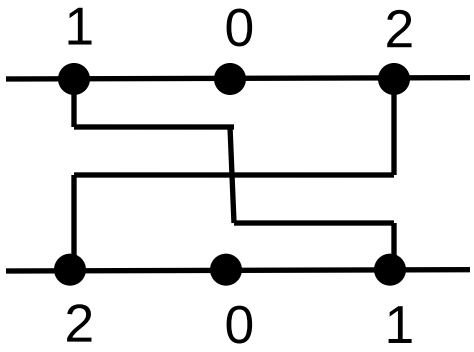


Vertical constraint graph



- Dogleg can solve the problem

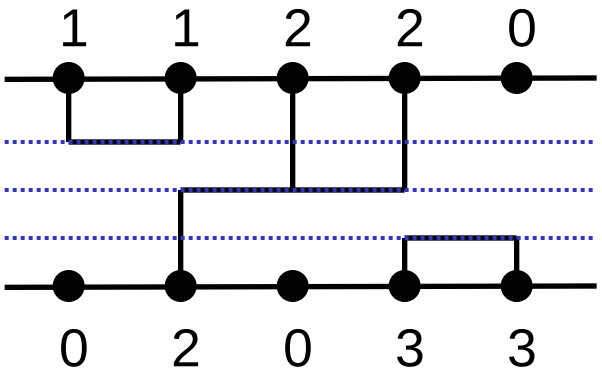




Reduce Channel Width by Dogleg

- Even without cycle in the VCG, Dogleg is useful because it can reduce channel width

Without dogleg



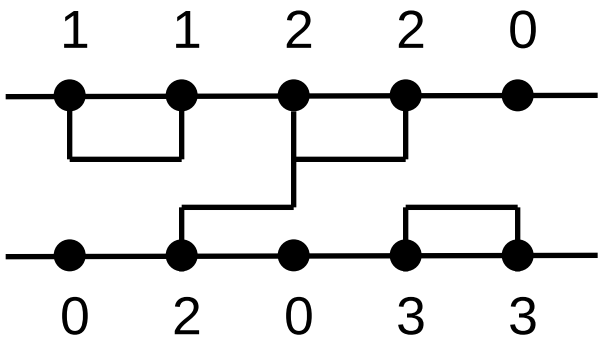
With dogleg

???

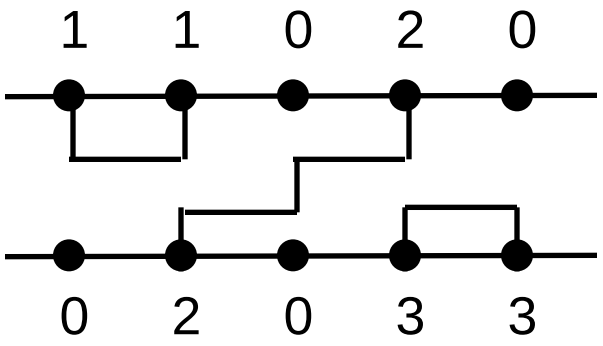
Deutch's Dogleg Algorithm

- Split each multi-terminal net into several horizontal segments
- Split only at columns that contain a pin of the net, i.e., using restricted dogleg only
- Apply the Constrained Left-edge algorithm

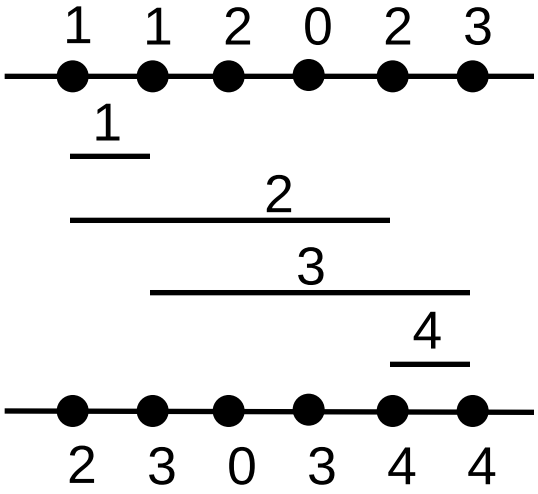
Restricted dogleg



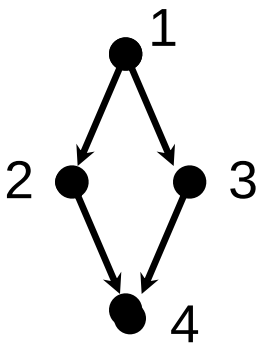
Unrestricted dogleg



Deutch's Dogleg Algorithm



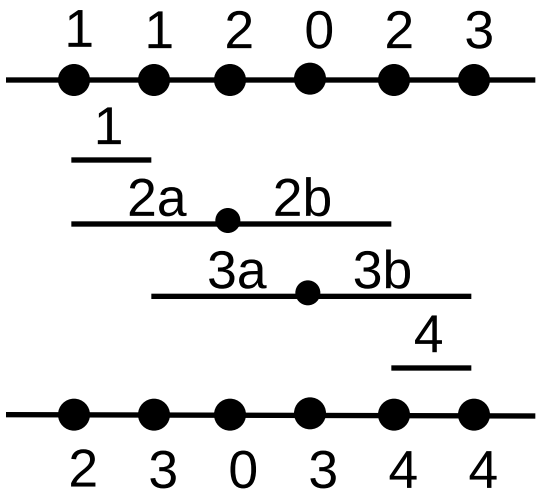
Vertical constraint graph



Routing without dogleg?

Vertical constraint graph after splitting?
Routing with dogleg?

Deutch's Dogleg Algorithm



Vertical constraint graph after splitting

???

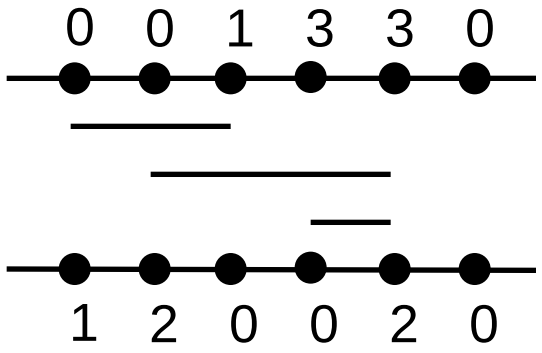
Without dogleg

???

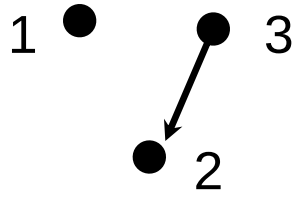
By Deutch's Dogleg alg.

???

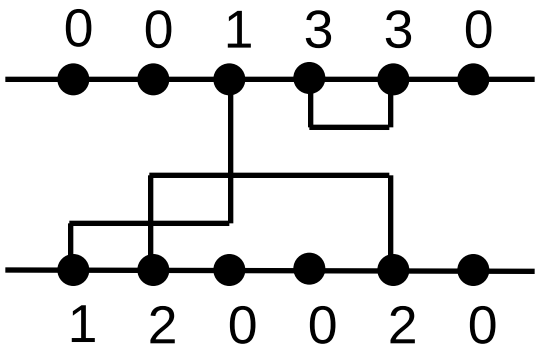
Drawbacks of the Constrained Left-edge Algorithm



Vertical constraint graph



By Constrained Left-edge algorithm



There is a better solution...

Drawbacks of the Constrained Left-edge Algorithm

- What's wrong with the Constrained Left-edge algorithm?
  - The Constrained Left-edge algorithm does not take care of the vertical and horizontal constraints together optimally

